

Machine Learning

Supervised and Unsupervised Learning

Supervised Learning

Supervised Learning is a machine learning approach that's defined by its use of labeled datasets. These datasets are designed to train or "supervise" algorithms into classifying data or predicting outcomes accurately. Using labeled inputs and outputs, the model can measure its accuracy and learn over time.

Supervised learning can be separated into two types of problems when data mining: classification and regression:

- **Classification** problems use an algorithm to accurately assign test data into specific categories, such as separating apples from oranges. Or, in the real world, supervised learning algorithms can be used to classify spam in a separate folder from your inbox. Linear classifiers, support vector machines, decision trees and random forest are all common types of classification algorithms.
- **Regression** is another type of supervised learning method that uses an algorithm to understand the relationship between dependent and independent variables. Regression models are helpful for predicting numerical values based on different data points, such as sales revenue projections for a given business. Some popular regression algorithms are linear regression, logistic regression and polynomial regression.

Evaluating Supervised Learning Models

Evaluating supervised learning models is an important step in ensuring that the model is accurate and generalizable. There are a number of different metrics that can be used to evaluate supervised learning models, but some of the most common ones include:

For Regression

- **Mean Squared Error (MSE):** MSE measures the average squared difference between the predicted values and the actual values. Lower MSE values indicate better model performance.
- **Root Mean Squared Error (RMSE):** RMSE is the square root of MSE, representing the standard deviation of the prediction errors. Similar to MSE, lower RMSE values indicate better model performance.
- **Mean Absolute Error (MAE):** MAE measures the average absolute difference between the predicted values and the actual values. It is less sensitive to outliers compared to MSE or RMSE.
- **R-squared (Coefficient of Determination):** R-squared measures the proportion of the variance in the target variable that is explained by the model. Higher R-squared values indicate better model fit.

For Classification

- **Accuracy:** Accuracy is the percentage of predictions that the model makes correctly. It is calculated by dividing the number of correct predictions by the total number of predictions.
- **Precision:** Precision is the percentage of positive predictions that the model makes that are actually correct. It is calculated by dividing the number of true positives by the total number of positive predictions.
- **Recall:** Recall is the percentage of all positive examples that the model correctly identifies. It is calculated by dividing the number of true positives by the total number of positive examples.

- **F1 score:** The F1 score is a weighted average of precision and recall. It is calculated by taking the harmonic mean of precision and recall.
- **Confusion matrix:** A confusion matrix is a table that shows the number of predictions for each class, along with the actual class labels. It can be used to visualize the performance of the model and identify areas where the model is struggling.

Unsupervised Learning

Unsupervised learning uses machine learning algorithms to analyze and cluster unlabeled data sets. These algorithms discover hidden patterns in data without the need for human intervention (hence, they are “unsupervised”).

Unsupervised learning models are used for three main tasks: clustering, association and dimensionality reduction:

- **Clustering** is a data mining technique for grouping unlabeled data based on their similarities or differences. For example, K-means clustering algorithms assign similar data points into groups, where the K value represents the size of the grouping and granularity. This technique is helpful for market segmentation, image compression, etc.
- **Association** is another type of unsupervised learning method that uses different rules to find relationships between variables in a given dataset. These methods are frequently used for market basket analysis and recommendation engines, along the lines of “Customers Who Bought This Item Also Bought” recommendations.
- **Dimensionality reduction** is a learning technique used when the number of features (or dimensions) in a given dataset is too high. It reduces the number of data inputs to a manageable size while also preserving the data integrity. Often, this technique is used in the preprocessing data stage, such as when autoencoders remove noise from visual data to improve picture quality.

Evaluating Unsupervised Learning Models

Evaluating non-supervised learning models is an important step in ensuring that the model is effective and useful. However, it can be more challenging than evaluating supervised learning models, as there is no ground truth data to compare the model’s predictions to. There are a number of different metrics that can be used to evaluate non-supervised learning models, but some of the most common ones include:

- **Silhouette score:** The silhouette score measures how well each data point is clustered with its own cluster members and separated from other clusters. It ranges from -1 to 1, with higher scores indicating better clustering.
- **Calinski-Harabasz score:** The Calinski-Harabasz score measures the ratio between the variance between clusters and the variance within clusters. It ranges from 0 to infinity, with higher scores indicating better clustering.
- **Adjusted Rand index:** The adjusted Rand index measures the similarity between two clustering’s. It ranges from -1 to 1, with higher scores indicating more similar clustering’s.
- **Davies-Bouldin index:** The Davies-Bouldin index measures the average similarity between clusters. It ranges from 0 to infinity, with lower scores indicating better clustering.
- **F1 score:** The F1 score is a weighted average of precision and recall, which are two metrics that are commonly used in supervised learning to evaluate classification models. However, the F1 score can also be used to evaluate non-supervised learning models, such as clustering models.

Classification

Classification is a supervised machine learning method where the model tries to predict the correct label of a given input data. In classification, the model is fully trained using the training data, and then it is evaluated on test data before being used to perform prediction on new unseen data.

Lazy Learners Vs. Eager Learners

There are two types of learners in machine learning classification: lazy and eager learners.

Eager learners are machine learning algorithms that first build a model from the training dataset before making any prediction on future datasets. They spend more time during the training process because of their eagerness to have a better generalization during the training from learning the weights, but they require less time to make predictions.

Most machine learning algorithms are eager learners, and below are some examples:

- Logistic Regression.
- Support Vector Machine.
- Decision Trees.
- Artificial Neural Networks.

Lazy learners or instance-based learners, on the other hand, do not create any model immediately from the training data, and this is where the lazy aspect comes from. They just memorize the training data, and each time there is a need to make a prediction, they search for the nearest neighbor from the whole training data, which makes them very slow during prediction. Some examples of this kind are:

- K-Nearest Neighbor.
- Case-based reasoning.

However, some algorithms, such as BallTrees and KDTrees, can be used to improve the prediction latency.

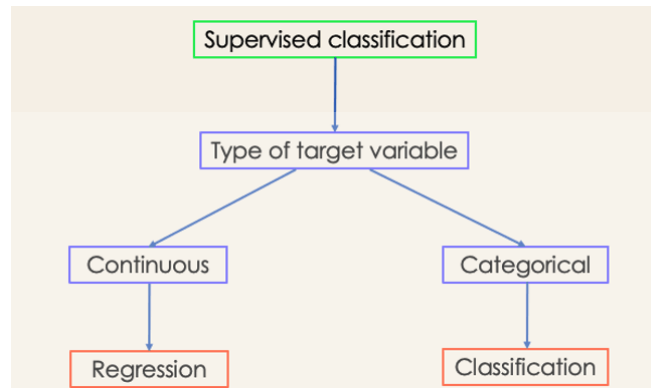
Machine Learning Classification Vs. Regression

There are four main categories of Machine Learning algorithms: supervised, unsupervised, semi-supervised, and reinforcement learning.

Even though classification and regression are both from the category of supervised learning, they are not the same.

- The prediction task is a classification when the target variable is discrete. An application is the identification of the underlying sentiment of a piece of text.

- The prediction task is a regression when the target variable is continuous. An example can be the prediction of the salary of a person given their education degree, previous work experience, geographical location, and level of seniority.

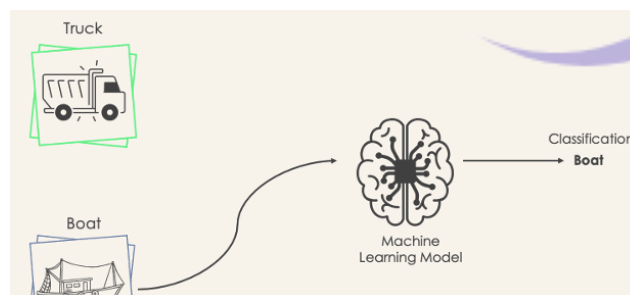


Types of Classification

There are four main classification tasks in Machine learning: binary, multi-class, multi-label, and imbalanced classifications.

Binary Classification

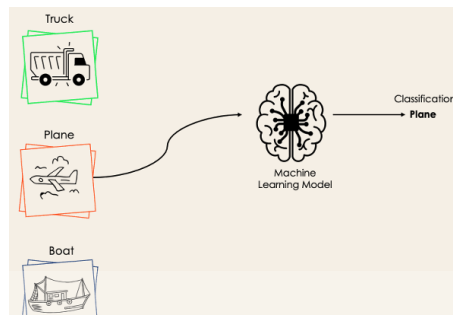
In a binary classification task, the goal is to classify the input data into two mutually exclusive categories. The training data in such a situation is labeled in a binary format: true and false; positive and negative; 0 and 1; spam and not spam, etc. depending on the problem being tackled. For instance, we might want to detect whether a given image is a truck or a boat.



Logistic Regression and Support Vector Machines algorithms are natively designed for binary classifications. However, other algorithms such as K-Nearest Neighbors and Decision Trees can also be used for binary classification.

Multi-Class Classification

The multi-class classification, on the other hand, has at least two mutually exclusive class labels, where the goal is to predict to which class a given input example belongs to. In the following case, the model correctly classified the image to be a plane.



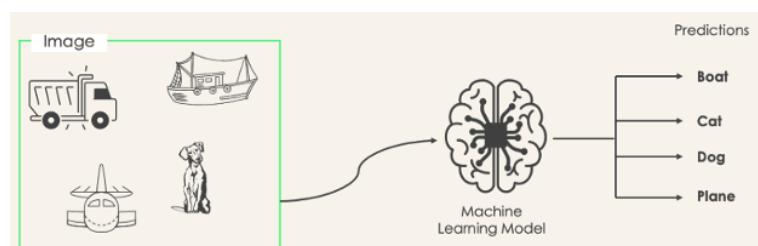
Most of the binary classification algorithms can be also used for multi-class classification. These algorithms include but are not limited to:

- Random Forest
- Naive Bayes
- K-Nearest Neighbors
- Gradient Boosting
- SVM
- Logistic Regression.

Multi-Label Classification

In multi-label classification tasks, we try to predict 0 or more classes for each input example. In this case, there is no mutual exclusion because the input example can have more than one label.

Such a scenario can be observed in different domains, such as auto-tagging in Natural Language Processing, where a given text can contain multiple topics. Similarly to computer vision, an image can contain multiple objects, as illustrated below: the model predicted that the image contains: a plane, a boat, a truck, and a dog.

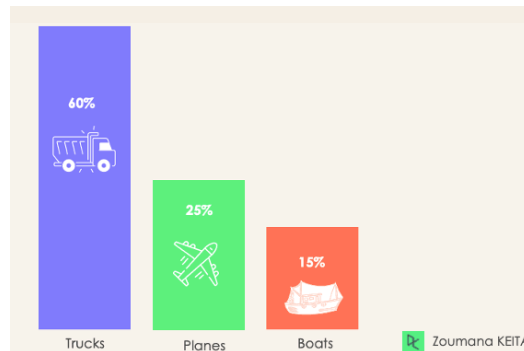


It is not possible to use multi-class or binary classification models to perform multi-label classification. However, most algorithms used for those standard classification tasks have their specialized versions for multi-label classification. We can cite:

- Multi-label Decision Trees
- Multi-label Gradient Boosting
- Multi-label Random Forests

Imbalanced Classification

For the imbalanced classification, the number of examples is unevenly distributed in each class, meaning that we can have more of one class than the others in the training data. Let's consider the following 3-class classification scenario where the training data contains: 60% of trucks, 25% of planes, and 15% of boats.



The imbalanced classification problem could occur in the following scenario:

- Fraudulent transaction detections in financial industries
- Rare disease diagnosis
- Customer churn analysis

Using conventional predictive models such as Decision Trees, Logistic Regression, etc. could not be effective when dealing with an imbalanced dataset, because they might be biased toward predicting the class with the highest number of observations, and considering those with fewer numbers as noise.

Metrics to Evaluate Machine Learning Classification Algorithms

Now that we have an idea of the different types of classification models, it is crucial to choose the right evaluation metrics for those models. In this section, we will cover the most commonly used metrics: accuracy, precision, recall, F1 score, and area under the ROC (Receiver Operating Characteristic) curve and AUC (Area Under the Curve).

A bit of context

Imagine you are a healthcare startup, and want an AI assistant able to predict whether a given patient has a heart disease or not based on its health record. This is a binary classification problem where the model will predict

- 1, True or Yes if the patient has heart disease
- 0, False or No otherwise

1 Confusion matrix

A 2X2 matrix that nicely summarizes the number of correct predictions of the model. It also helps in computing different other performance metrics.

Predicti Reality	Yes	No
Yes	True Positives (TP)	False Negatives (FN)
No	False Positives (FP)	True Negatives (TN)

Type I Error

Type II Error

Type I & II Errors can be used interchangeably when referring to False Positives and False negatives respectively

2 Accuracy

We get accuracy by answering this question: "out of the predictions made by the model, what percentage is correct?"

$$\text{Accuracy} = \frac{TP + TN}{\text{Total number observation}}$$

3 Precision

We get precision by answering this question: "out of all the YES predictions, how many of them were correct?"

$$\text{Precision} = \frac{TP}{TP + FP}$$

4 Recall / Sensitivity

It aims to answer this question: "how good was the model at predicting real Yes events?", which can be considered as the flip of the precision.

$$\text{Sensitivity} = \frac{TP}{TP + FN}$$

5 Recall / Specificity

It aims to answer this question: "how good was the model at predicting real No events?".

$$\text{Specificity} = \frac{TN}{TN + FP}$$

6 F1 Score

Sometimes used when dealing with imbalanced data set, meaning that there are more of one class/label than there are of the other. It corresponds to the harmonic mean of the precision and recall.

$$\text{F1 Score} = 2 \cdot \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

7 AUC – ROC Curve

AUC– ROC generates probability values instead of binary 0/1 values. It should be used when your data set is roughly balanced.

Using ROC for imbalanced data sets lead to incorrect interpretation.

ROC curves provide good overview of trade-off between the TP rate and FP rate for binary classifier using different probability thresholds.

- A value below 0.5 indicates a poor classifier
- A value of 0.5 means random classifier
- Value over 0.7 corresponds to a good classifier
- 0.8 indicates a strong classifier
- We have 1 when the classifier perfectly predicts everything.

Strategies to choose the right metric

Choose accuracy

- The cost of FP and FN are roughly equal.
- The benefit of TP and TN are roughly equal.

Choose Precision

- The cost of FP is much higher than a FN.
- The benefit of a TP is much higher than a TN.

Choose recall

- The cost of FN is much higher than a FP.
- The cost of a TN is much higher than a TP.

ROC AUC & Precision – Recall curves

- Use ROC when the dealing with balanced data sets.
- Use precision-recall for imbalanced data sets.

Decision Tree Learning

A decision tree is a type of supervised learning algorithm that is commonly used in machine learning to model and predict outcomes based on input data. It is a tree-like structure where each internal node tests on attribute, each branch corresponds to attribute value and each leaf node represents the final decision or prediction. The decision tree algorithm falls under the category of supervised learning. They can be used to solve both regression and classification problems.

Decision Tree Terminologies

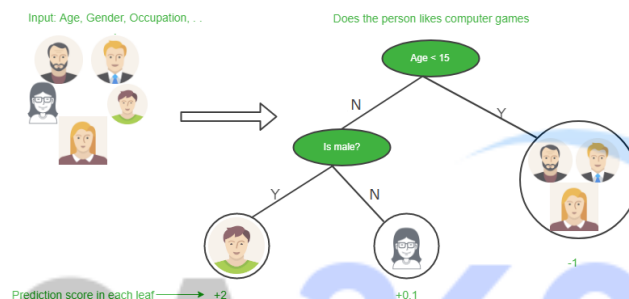
There are specialized terms associated with decision trees that denote various components and facets of the tree structure and decision-making procedure. :

- **Root Node:** A decision tree's root node, which represents the original choice or feature from which the tree branches, is the highest node.
- **Internal Nodes (Decision Nodes):** Nodes in the tree whose choices are determined by the values of particular attributes. There are branches on these nodes that go to other nodes.
- **Leaf Nodes (Terminal Nodes):** The branches' termini, when choices or forecasts are decided upon. There are no more branches on leaf nodes.
- **Branches (Edges):** Links between nodes that show how decisions are made in response to particular circumstances.

- **Splitting:** The process of dividing a node into two or more sub-nodes based on a decision criterion. It involves selecting a feature and a threshold to create subsets of data.
- **Parent Node:** A node that is split into child nodes. The original node from which a split originates.
- **Child Node:** Nodes created as a result of a split from a parent node.
- **Decision Criterion:** The rule or condition used to determine how the data should be split at a decision node. It involves comparing feature values against a threshold.
- **Pruning:** The process of removing branches or nodes from a decision tree to improve its generalization and prevent overfitting.

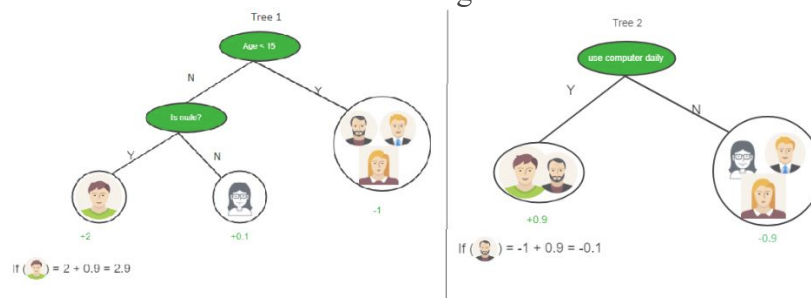
Decision Tree Approach

Decision tree uses the tree representation to solve the problem in which each leaf node corresponds to a class label and attributes are represented on the internal node of the tree. We can represent any Boolean function on discrete attributes using the decision tree.



Below are some assumptions that we made while using the decision tree:
At the beginning, we consider the whole training set as the root.

- Feature values are preferred to be categorical. If the values are continuous then they are discretized prior to building the model.
- On the basis of attribute values, records are distributed recursively.
- We use statistical methods for ordering attributes as root or the internal node.



Attribute Selection Measures

While implementing a Decision tree, the main issue arises that how to select the best attribute for the root node and for sub-nodes. So, to solve such problems there is a technique which is called as Attribute selection measure or ASM. By this measurement, we can easily select the best attribute for the nodes of the tree. There are two popular techniques for ASM, which are:

- Information Gain

- Gini Index

1. Information Gain:

- Information gain is the measurement of changes in entropy after the segmentation of a dataset based on an attribute.
- It calculates how much information a feature provides us about a class.
- According to the value of information gain, we split the node and build the decision tree.
- A decision tree algorithm always tries to maximize the value of information gain, and a node/attribute having the highest information gain is split first. It can be calculated using the below formula:

$$1. \text{ Information Gain} = \text{Entropy}(S) - [(\text{Weighted Avg}) * \text{Entropy (each feature)}]$$

Entropy: Entropy is a metric to measure the impurity in a given attribute. It specifies randomness in data. Entropy can be calculated as:

$$\text{Entropy}(s) = -P(\text{yes}) \log_2 P(\text{yes}) - P(\text{no}) \log_2 P(\text{no})$$

Where,

- S = Total number of samples
- $P(\text{yes})$ = probability of yes
- $P(\text{no})$ = probability of no

2. Gini Index:

- Gini index is a measure of impurity or purity used while creating a decision tree in the CART (Classification and Regression Tree) algorithm.
- An attribute with the low Gini index should be preferred as compared to the high Gini index.
- It only creates binary splits, and the CART algorithm uses the Gini index to create binary splits.
- Gini index can be calculated using the below formula:

$$\text{Gini Index} = 1 - \sum_j P_j^2$$

Advantages of the Decision Tree

- It is simple to understand as it follows the same process which a human follow while making any decision in real-life.
- It can be very useful for solving decision-related problems.
- It helps to think about all the possible outcomes for a problem.
- There is less requirement of data cleaning compared to other algorithms.

Disadvantages of the Decision Tree

- The decision tree contains lots of layers, which makes it complex.
- It may have an overfitting issue, which can be resolved using the **Random Forest algorithm**.
- For more class labels, the computational complexity of the decision tree may increase.

CART (Classification and Regression Tree)

CART (Classification and Regression Trees) is a variation of the decision tree algorithm. It can handle both classification and regression tasks.

CART (Classification and Regression Tree) for Decision Tree

CART is a predictive algorithm used in Machine learning and it explains how the target variable's values can be predicted based on other matters. It is a decision tree where each fork is split into a predictor variable and each node has a prediction for the target variable at the end.

The term CART serves as a generic term for the following categories of decision trees:

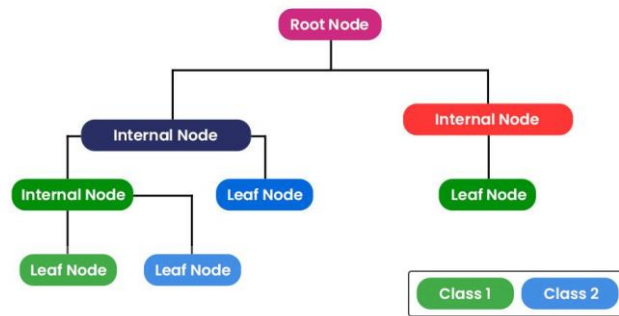
- **Classification Trees:** The tree is used to determine which “class” the target variable is most likely to fall into when it is continuous.
- **Regression trees:** These are used to predict a continuous variable's value.

In the decision tree, nodes are split into sub-nodes based on a threshold value of an attribute. The root node is taken as the training set and is split into two by considering the best attribute and threshold value. Further, the subsets are also split using the same logic. This continues till the last pure sub-set is found in the tree or the maximum number of leaves possible in that growing tree.

How does CART algorithm work?

The CART algorithm works via the following process:

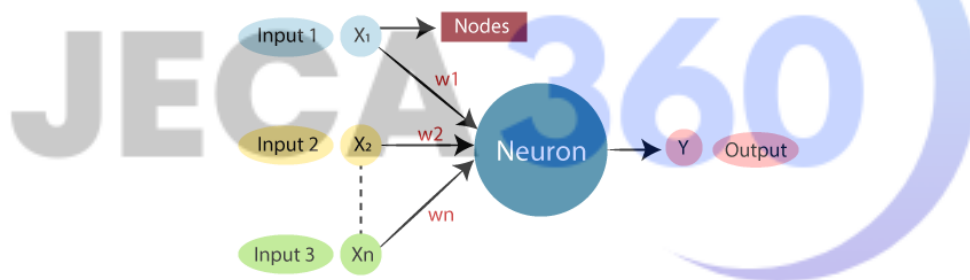
- The best-split point of each input is obtained.
- Based on the best-split points of each input in Step 1, the new “best” split point is identified.
- Split the chosen input according to the “best” split point.
- Continue splitting until a stopping rule is satisfied or no further desirable splitting is available.



CART algorithm uses Gini Impurity to split the dataset into a decision tree .It does that by searching for the best homogeneity for the sub nodes, with the help of the Gini index criterion.

Artificial Neural Network

The term "**Artificial Neural Network**" is derived from Biological neural networks that develop the structure of a human brain. Similar to the human brain that has neurons interconnected to one another, artificial neural networks also have neurons that are interconnected to one another in various layers of the networks. These neurons are known as nodes.

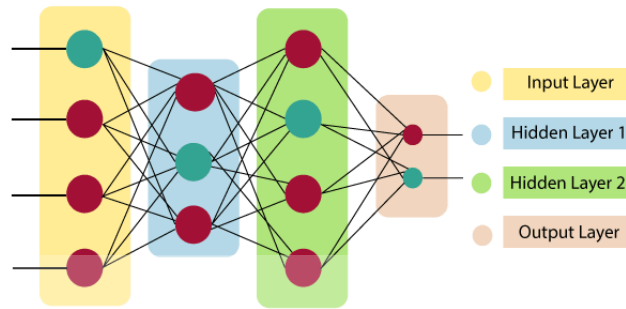


An **Artificial Neural Network** in the field of **Artificial intelligence** where it attempts to mimic the network of neurons makes up a human brain so that computers will have an option to understand things and make decisions in a human-like manner. The artificial neural network is designed by programming computers to behave simply like interconnected brain cells.

The architecture of an artificial neural network

To understand the concept of the architecture of an artificial neural network, we have to understand what a neural network consists of. In order to define a neural network that consists of a large number of artificial neurons, which are termed units arranged in a sequence of layers. Let's us look at various types of layers available in an artificial neural network.

Artificial Neural Network primarily consists of three layers:



Input Layer:

As the name suggests, it accepts inputs in several different formats provided by the programmer.

Hidden Layer:

The hidden layer presents in-between input and output layers. It performs all the calculations to find hidden features and patterns.

Output Layer:

The input goes through a series of transformations using the hidden layer, which finally results in output that is conveyed using this layer.

The artificial neural network takes input and computes the weighted sum of the inputs and includes a bias. This computation is represented in the form of a transfer function.

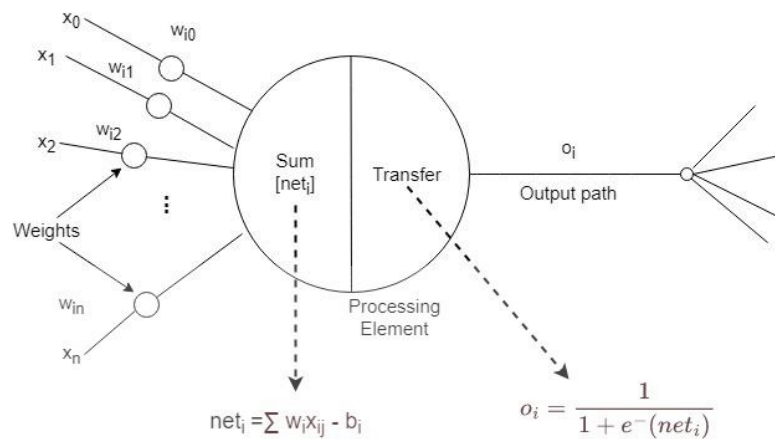
$$\sum_{i=1}^n W_i * X_i + b$$

It determines weighted total is passed as an input to an activation function to produce the output. Activation functions choose whether a node should fire or not. Only those who are fired make it to the output layer. There are distinctive activation functions available that can be applied upon the sort of task we are performing.

Neurons in ANN

An artificial neuron or neural node is a mathematical model. In most cases, it computes the weighted average of its input and then applies a bias to it. Post that, it passes this resultant term through an activation function. This activation function is a nonlinear function such as the sigmoid function that accepts a linear input and gives a nonlinear output.

The following figure shows a typical artificial neuron:



Neurons

A typical neural network consists of layers of neurons called neural nodes. These layers are of the following three types:

1. input layer (single)
2. hidden layer (one or more than one)
3. output layer (single)

Each neural node is connected to another and is characterized by its weight and a threshold. It gets an input on which it does some transformation and post that, it sends an output. If the output of any individual node is above the specified threshold value, that node gets activated. Then, it sends data to the next layer of the network. Otherwise, it remains dormant and thus doesn't transmit any data to the next layer of the network.

Types of Artificial Neural Network:

There are various types of Artificial Neural Networks (ANN) depending upon the human brain neuron and network functions, an artificial neural network similarly performs tasks. The majority of the artificial neural networks will have some similarities with a more complex biological partner and are very effective at their expected tasks. For example, segmentation or classification.

Feedback ANN:

In this type of ANN, the output returns into the network to accomplish the best-evolved results internally. As per the University of Massachusetts, Lowell Centre for Atmospheric Research. The feedback networks feed information back into itself and are well suited to solve optimization issues. The Internal system error corrections utilize feedback ANNs.

Feed-Forward ANN:

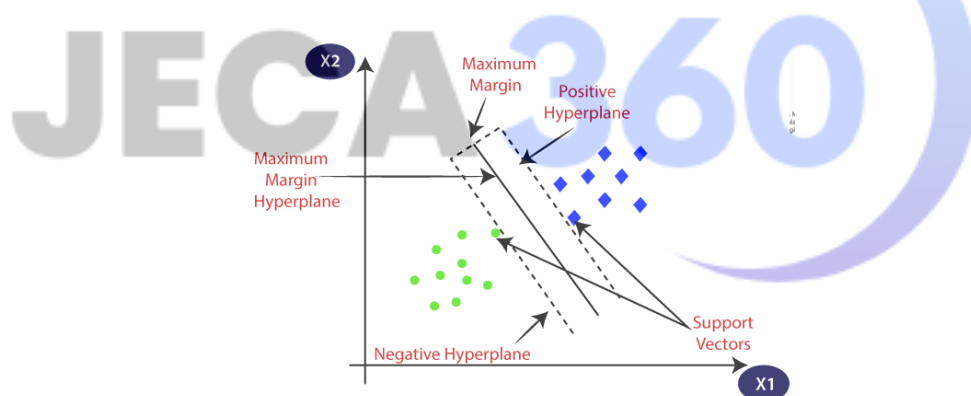
A feed-forward network is a basic neural network comprising of an input layer, an output layer, and at least one layer of a neuron. Through assessment of its output by reviewing its input, the intensity of the network can be noticed based on group behavior of the associated neurons, and the output is decided. The primary advantage of this network is that it figures out how to evaluate and recognize input patterns.

Support Vector Machines

Support Vector Machine or SVM is one of the most popular Supervised Learning algorithms, which is used for Classification as well as Regression problems. However, primarily, it is used for Classification problems in Machine Learning.

The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that we can easily put the new data point in the correct category in the future. This best decision boundary is called a hyperplane.

SVM chooses the extreme points/vectors that help in creating the hyperplane. These extreme cases are called as support vectors, and hence algorithm is termed as Support Vector Machine. Consider the below diagram in which there are two different categories that are classified using a decision boundary or hyperplane:



Example: SVM can be understood with the example that we have used in the KNN classifier. Suppose we see a strange cat that also has some features of dogs, so if we want a model that can accurately identify whether it is a cat or dog, so such a model can be created by using the SVM algorithm. We will first train our model with lots of images of cats and dogs so that it can learn about different features of cats and dogs, and then we test it with this strange creature. So as support vector creates a decision boundary between these two data (cat and dog) and choose extreme cases (support vectors), it will see the extreme case of cat and dog. On the basis of the support vectors, it will classify it as a cat.

SVM algorithm can be used for **Face detection, image classification, text categorization, etc.**

Types of SVM

SVM can be of two types:

- **Linear SVM:** Linear SVM is used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as Linear SVM classifier.
- **Non-linear SVM:** Non-Linear SVM is used for non-linearly separated data, which means if a dataset cannot be classified by using a straight line, then such data is termed as non-linear data and classifier used is called as Non-linear SVM classifier.

Hyperplane and Support Vectors in the SVM algorithm:

Hyperplane: There can be multiple lines/decision boundaries to segregate the classes in n-dimensional space, but we need to find out the best decision boundary that helps to classify the data points. This best boundary is known as the hyperplane of SVM.

The dimensions of the hyperplane depend on the features present in the dataset, which means if there are 2 features (as shown in image), then hyperplane will be a straight line. And if there are 3 features, then hyperplane will be a 2-dimension plane.

We always create a hyperplane that has a maximum margin, which means the maximum distance between the data points.

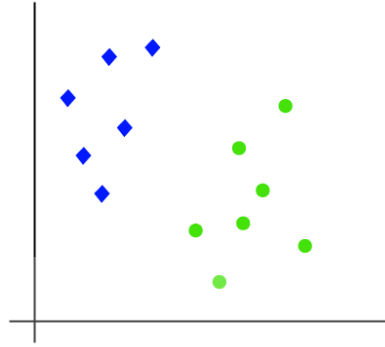
Support Vectors:

The data points or vectors that are the closest to the hyperplane and which affect the position of the hyperplane are termed as Support Vector. Since these vectors support the hyperplane, hence called a Support vector.

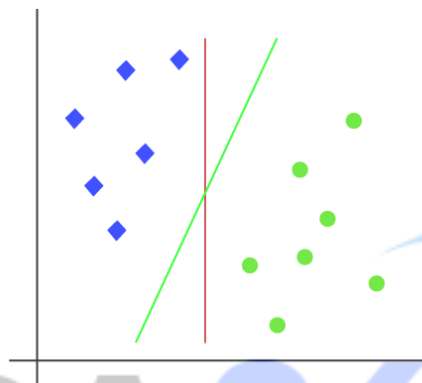
How does SVM works?

Linear SVM:

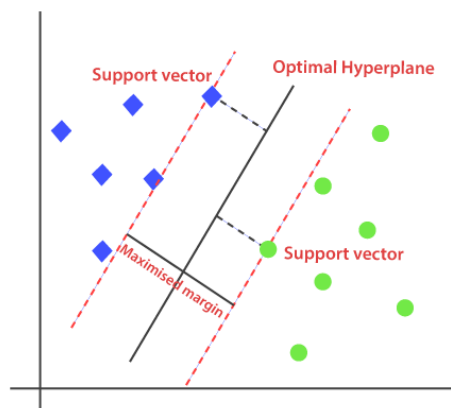
The working of the SVM algorithm can be understood by using an example. Suppose we have a dataset that has two tags (green and blue), and the dataset has two features x_1 and x_2 . We want a classifier that can classify the pair (x_1, x_2) of coordinates in either green or blue. Consider the below image:



So, as it is 2-d space so by just using a straight line, we can easily separate these two classes. But there can be multiple lines that can separate these classes. Consider the below image:

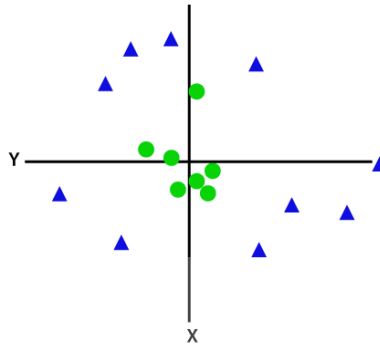


Hence, the SVM algorithm helps to find the best line or decision boundary; this best boundary or region is called as a **hyperplane**. SVM algorithm finds the closest point of the lines from both the classes. These points are called support vectors. The distance between the vectors and the hyperplane is called as **margin**. And the goal of SVM is to maximize this margin. The **hyperplane** with maximum margin is called the **optimal hyperplane**.



Non-Linear SVM:

If data is linearly arranged, then we can separate it by using a straight line, but for non-linear data, we cannot draw a single straight line. Consider the below image:



So, to separate these data points, we need to add one more dimension. For linear data, we have used two dimensions x and y , so for non-linear data, we will add a third-dimension z . It can be calculated as:

$$z = x^2 + y^2$$

Hence, we get a circumference of radius 1 in case of non-linear data.

Bayesian Learning

Bayes theorem is also known with some other name such as **Bayes rule or Bayes Law**. **Bayes theorem helps to determine the probability of an event with random knowledge**. It is used to calculate the probability of occurring one event while other one already occurred. It is a best method to relate the condition probability and marginal probability.

In simple words, we can say that Bayes theorem helps to contribute more accurate results.

Bayes Theorem is used to estimate the precision of values and provides a method for calculating the conditional probability. However, it is hypocritically a simple calculation but it is used to easily calculate the conditional probability of events where intuition often fails. Some of the data scientist assumes that Bayes theorem is most widely used in financial industries but it is not like that. Other than financial, Bayes theorem is also extensively applied in health and medical, research and survey industry, aeronautical sector, etc.

Bayes Theorem

Bayes theorem is one of the most popular machine learning concepts that helps to calculate the probability of occurring one event with uncertain knowledge while other one has already occurred.

Bayes' theorem can be derived using product rule and conditional probability of event X with known event Y :

- According to the product rule we can express as the probability of event X with known event Y as follows;

1. $P(X \text{ ? } Y) = P(X|Y) P(Y)$ {equation 1}

- Further, the probability of event Y with known event X:

$$1. P(X \rightarrow Y) = P(Y|X) P(X) \quad \{\text{equation 2}\}$$

Mathematically, Bayes theorem can be expressed by combining both equations on right hand side. We will get:

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$

Here, both events X and Y are independent events which means probability of outcome of both events does not depends one another.

The above equation is called as Bayes Rule or Bayes Theorem.

- $P(X|Y)$ is called as **posterior**, which we need to calculate. It is defined as updated probability after considering the evidence.
- $P(Y|X)$ is called the likelihood. It is the probability of evidence when hypothesis is true.
- $P(X)$ is called the **prior probability**, probability of hypothesis before considering the evidence
- $P(Y)$ is called marginal probability. It is defined as the probability of evidence under any consideration.

Hence, Bayes Theorem can be written as:

$$\text{posterior} = \text{likelihood} * \text{prior} / \text{evidence}$$

Naïve Bayes Classifier Algorithm

- Naïve Bayes algorithm is a supervised learning algorithm, which is based on **Bayes theorem** and used for solving classification problems.
- It is mainly used in *text classification* that includes a high-dimensional training dataset.
- Naïve Bayes Classifier is one of the simple and most effective Classification algorithms which helps in building the fast machine learning models that can make quick predictions.
- **It is a probabilistic classifier, which means it predicts on the basis of the probability of an object.**
- Some popular examples of Naïve Bayes Algorithm are **spam filtration, Sentimental analysis, and classifying articles.**

Why is it called Naïve Bayes?

The Naïve Bayes algorithm is comprised of two words Naïve and Bayes, which can be described as:

- **Naïve:** It is called Naïve because it assumes that the occurrence of a certain feature is independent of the occurrence of other features. Such as if the fruit is identified on the bases of color, shape, and taste, then red, spherical, and sweet fruit is recognized as an apple. Hence each feature individually contributes to identify that it is an apple without depending on each other.
- **Bayes:** It is called Bayes because it depends on the principle of Bayes' Theorem.

Clustering

Clustering or cluster analysis is a machine learning technique, which groups the unlabeled dataset. It can be defined as *"A way of grouping the data points into different clusters, consisting of similar data points. The objects with the possible similarities remain in a group that has less or no similarities with another group."*

It does it by finding some similar patterns in the unlabeled dataset such as shape, size, color, behavior, etc., and divides them as per the presence and absence of those similar patterns.

It is an unsupervised learning method, hence no supervision is provided to the algorithm, and it deals with the unlabeled dataset.

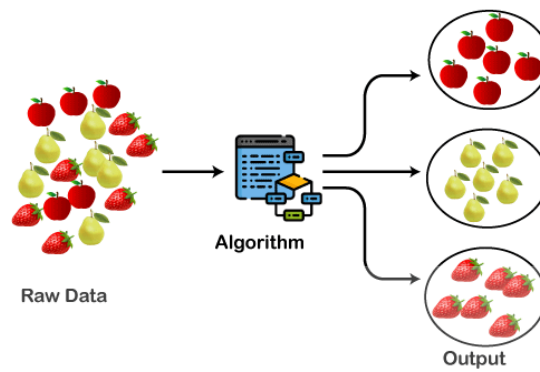
After applying this clustering technique, each cluster or group is provided with a cluster-ID. ML system can use this id to simplify the processing of large and complex datasets.

The clustering technique is commonly used for **statistical data analysis**.

Note: Clustering is somewhere similar to the classification algorithm, but the difference is the type of dataset that we are using. In classification, we work with the labeled data set, whereas in clustering, we work with the unlabeled dataset.

Example: Let's understand the clustering technique with the real-world example of Mall: When we visit any shopping mall, we can observe that the things with similar usage are grouped together. Such as the t-shirts are grouped in one section, and trousers are at other sections, similarly, at vegetable sections, apples, bananas, Mangoes, etc., are grouped in separate sections, so that we can easily find out the things. The clustering technique also works in the same way. Other examples of clustering are grouping documents according to the topic.

The below diagram explains the working of the clustering algorithm. We can see the different fruits are divided into several groups with similar properties.



Types of Clustering Methods

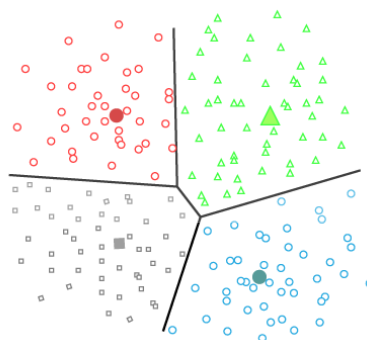
The clustering methods are broadly divided into **Hard clustering** (datapoint belongs to only one group) and **Soft Clustering** (data points can belong to another group also). But there are also other various approaches of Clustering exist. Below are the main clustering methods used in Machine learning:

1. **Partitioning Clustering**
2. **Density-Based Clustering**
3. **Distribution Model-Based Clustering**
4. **Hierarchical Clustering**
5. **Fuzzy Clustering**

Partitioning Clustering

It is a type of clustering that divides the data into non-hierarchical groups. It is also known as the **centroid-based method**. The most common example of partitioning clustering is the **K-Means Clustering algorithm**.

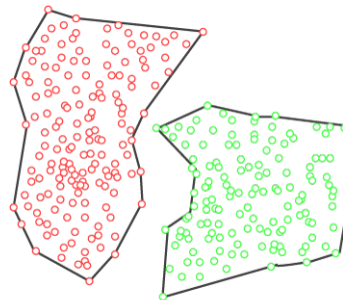
In this type, the dataset is divided into a set of k groups, where K is used to define the number of pre-defined groups. The cluster center is created in such a way that the distance between the data points of one cluster is minimum as compared to another cluster centroid.



Density-Based Clustering

The density-based clustering method connects the highly-dense areas into clusters, and the arbitrarily shaped distributions are formed as long as the dense region can be connected. This algorithm does it by identifying different clusters in the dataset and connects the areas of high densities into clusters. The dense areas in data space are divided from each other by sparser areas.

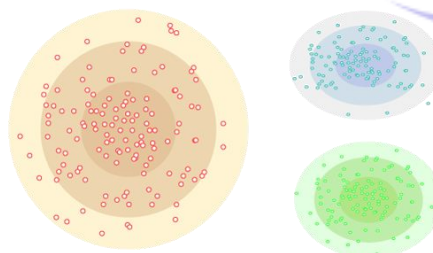
These algorithms can face difficulty in clustering the data points if the dataset has varying densities and high dimensions.



Distribution Model-Based Clustering

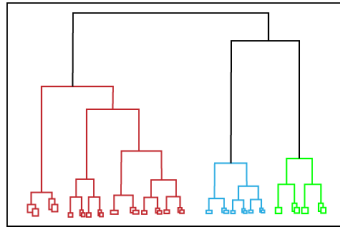
In the distribution model-based clustering method, the data is divided based on the probability of how a dataset belongs to a particular distribution. The grouping is done by assuming some distributions commonly **Gaussian Distribution**.

The example of this type is the **Expectation-Maximization Clustering algorithm** that uses Gaussian Mixture Models (GMM).



Hierarchical Clustering

Hierarchical clustering can be used as an alternative for the partitioned clustering as there is no requirement of pre-specifying the number of clusters to be created. In this technique, the dataset is divided into clusters to create a tree-like structure, which is also called a **dendrogram**. The observations or any number of clusters can be selected by cutting the tree at the correct level. The most common example of this method is the **Agglomerative Hierarchical algorithm**.



Fuzzy Clustering

Fuzzy clustering is a type of soft method in which a data object may belong to more than one group or cluster. Each dataset has a set of membership coefficients, which depend on the degree of membership to be in a cluster. **Fuzzy C-means algorithm** is the example of this type of clustering; it is sometimes also known as the Fuzzy k-means algorithm.

Clustering Algorithms

The Clustering algorithms can be divided based on their models that are explained above. There are different types of clustering algorithms published, but only a few are commonly used. The clustering algorithm is based on the kind of data that we are using. Such as, some algorithms need to guess the number of clusters in the given dataset, whereas some are required to find the minimum distance between the observation of the dataset.

Here we are discussing mainly popular Clustering algorithms that are widely used in machine learning:

1. **K-Means algorithm:** The k-means algorithm is one of the most popular clustering algorithms. It classifies the dataset by dividing the samples into different clusters of equal variances. The number of clusters must be specified in this algorithm. It is fast with fewer computations required, with the linear complexity of $O(n)$.
2. **Mean-shift algorithm:** Mean-shift algorithm tries to find the dense areas in the smooth density of data points. It is an example of a centroid-based model, that works on updating the candidates for centroid to be the center of the points within a given region.
3. **DBSCAN Algorithm:** It stands for **Density-Based Spatial Clustering of Applications with Noise**. It is an example of a density-based model similar to the mean-shift, but with some remarkable advantages. In this algorithm, the areas of high density are separated by the areas of low density. Because of this, the clusters can be found in any arbitrary shape.
4. **Expectation-Maximization Clustering using GMM:** This algorithm can be used as an alternative for the k-means algorithm or for those cases where K-means can be failed. In GMM, it is assumed that the data points are Gaussian distributed.
5. **Agglomerative Hierarchical algorithm:** The Agglomerative hierarchical algorithm performs the bottom-up hierarchical clustering. In this, each data point is treated as a

single cluster at the outset and then successively merged. The cluster hierarchy can be represented as a tree-structure.

6. **Affinity Propagation:** It is different from other clustering algorithms as it does not require to specify the number of clusters. In this, each data point sends a message between the pair of data points until convergence. It has $O(N^2T)$ time complexity, which is the main drawback of this algorithm.

Applications of Clustering

Below are some commonly known applications of clustering technique in Machine Learning:

- **In Identification of Cancer Cells**
- **In Search Engines**
- **Customer Segmentation**
- **In Biology**
- **In Land Use**

Hidden Markov Model

A Hidden Markov Model (HMM) is a probabilistic model that consists of a sequence of hidden states, each of which generates an observation. The hidden states are usually not directly observable, and the goal of HMM is to estimate the sequence of hidden states based on a sequence of observations. An HMM is defined by the following components:

- A set of N hidden states, $S = \{s_1, s_2, \dots, s_N\}$.
- A set of M observations, $O = \{o_1, o_2, \dots, o_M\}$.
- An initial state probability distribution, $\pi = \{\pi_1, \pi_2, \dots, \pi_N\}$, which specifies the probability of starting in each hidden state.
- A transition probability matrix, $A = [a_{ij}]$, defines the probability of moving from one hidden state to another.
- An emission probability matrix, $B = [b_{jk}]$, defines the probability of emitting an observation from a given hidden state.

The basic idea behind an HMM is that the hidden states generate the observations, and the observed data is used to estimate the hidden state sequence. This is often referred to as the forward-backwards algorithm.

An HMM consists of two types of variables: hidden states and observations.

- The **hidden states** are the underlying variables that generate the observed data, but they are not directly observable.
- The **observations** are the variables that are measured and observed.

The relationship between the hidden states and the observations is modeled using a probability distribution. The Hidden Markov Model (HMM) is the relationship between the hidden states and the observations using two sets of probabilities: the transition probabilities and the emission probabilities.

- The **transition probabilities** describe the probability of transitioning from one hidden state to another.
- The **emission probabilities** describe the probability of observing an output given a hidden state.

Distance Measures

Several machine learning techniques include distance measures as a significant component. These distance metrics are used to calculate the similarity between data points in both supervised and unsupervised learning.

Whether it's for classification or clustering, and efficient distance measure improves the performance of our machine learning model. In machine learning, there are four different types of distance metrics.

1. Euclidean Distance

The smallest distance between two points is known as the **Euclidean Distance**. This distance metric is used by most machine learning algorithms, including K-Means, to measure the similarity of observations. Assume we have the following two points:

$$P1 = (x_1, y_1)$$

$$P2 = (x_2, y_2)$$

Then, the formula for Euclidean distance can be expressed as -

$$d = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$$

We can generalize this for n-dimensional space as -

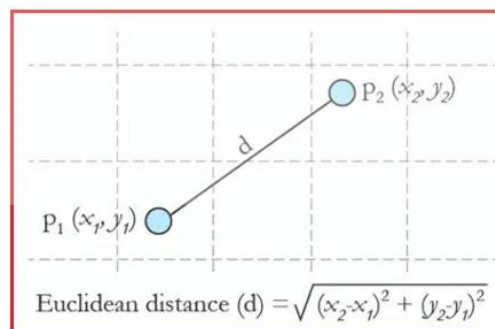
$$D_e = \left(\sum_{i=1}^n (x_i - y_i)^2 \right)^{\frac{1}{2}}$$

Where,

n = number of dimensions

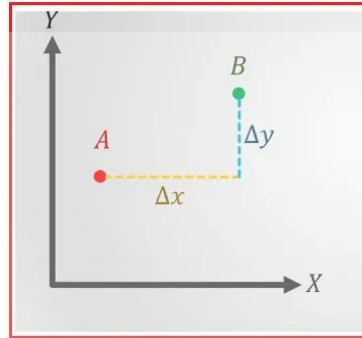
x_i, y_i = data points

Euclidean distance can be visualized graphically as –



2. Manhattan Distance

The Manhattan Distance is the total difference between two places in all dimensions. The term “Manhattan Distance” is frequently used to refer to the distance between two city blocks. Let’s consider two points – A = () and B = (). Manhattan Distance can be expressed graphically as follows:



Because the following illustration is two-dimensional, we’ll use the total of absolute distances in both the x and y axes to calculate Manhattan Distance.

In a two-dimensional space, the Manhattan distance is given as:

$$d = |x_1 - x_2| + |y_1 - y_2|$$

For an n-dimensional space, we can generalize the same as:

$$D_m = \sum_{i=1}^n |x_i - y_i|$$

Where,

n = number of dimensions

= data points

3. Minkowski Distance

The generalized form of the Euclidean and Manhattan Distances is the Minkowski Distance.

You can express the Minkowski distance as –

$$D = \left(\sum_{i=1}^n |x_i - y_i|^p \right)^{\frac{1}{p}}$$

The order of the norm is represented by p.

When an order(p) is 1, Manhattan Distance is represented, and when order(p) is 2 in the above formula, Euclidean Distance is represented.

4. Hamming Distance

The Hamming Distance is a measurement of how similar two strings of the same length are. The Hamming Distance is the number of spots where the corresponding characters differ between two strings of the same length.

References

- [1] <https://www.datacamp.com>
- [2] <https://www.javatpoint.com/>
- [3] <https://www.geeksforgeeks.org>
- [4] <https://www.baeldung.com>

